

Color Clash — Product Requirements Document

Rebuild v2.0 • React PWA • Online Accounts & Realtime Multiplayer

Dhoola • Geeks for Games

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1. Document control

Field	Value
Document	Product Requirements Document (PRD)
Product	Color Clash
Release	v2.0 — “Rebuild”
Status	Draft for approval
Supersedes	docs/MASTER_BLUEPRINT.md \$1, \$3, \$13, \$14 (v1.0)
Owner	Dhoola (Geeks for Games)
Companion documents	Technical Requirements Document (TRD), UI/UX Design Document, System Architecture & Backend Schema

1.1 What this document is

This PRD defines **what** Color Clash v2.0 must do and **why**. It does not prescribe implementation; that is the TRD’s job. Where v2.0 keeps a v1.0 behaviour unchanged, this document says so explicitly rather than restating it, so that “not mentioned” never means “silently dropped”.

1.2 Why a rebuild

Color Clash v1.0 shipped a genuinely good core and a deliberately minimal shell around it. The audit that preceded this document found the following.

What is worth keeping, essentially as-is:

- A pure, deterministic rule engine — `reduce(state, command, rng) → { state, events }` — with no clocks, I/O or globals, backing local play, the server, the bots and every test from one code path.
- Five isolated ruleset modules registered through a single `VersionManifest` interface, with deck compositions asserted at generation time.
- 212 unit/integration tests and 75 browser tests across three viewports.
- A documented rule-flag ledger (`RULE_FLAGS.md`, RF-001...RF-014) recording every place the source specification was silent and what was chosen instead.
- A completed rebrand off third-party artwork: all 41 game images generated from original vector sources, card faces drawn in CSS, all audio synthesised at runtime, with a CI audit that fails the build if the previous mark reappears.

What must be rebuilt:

- **There is no backend.** There is a single Node process with no database, no accounts, no authentication of any kind, and an in-memory room registry. `HELLO` accepts whatever `playerId` a client claims; sending someone else’s id takes over their seat.
- **Nothing is durable.** Room files are deleted 90 seconds after the last human leaves, and boot discards anything older than two minutes. A completed match leaves no trace anywhere. There is no match history, no stats, no leaderboard, no rating.
- **The economy is fiction.** Coins, gems and level live in `localStorage` with a starting balance of 12,450 coins and 860 gems, are client-authored, and are trivially edited.
- **There is no matchmaking.** The only route into a match is creating a room or typing a five-character code. No queue, no friends, no lobby browser.

- **It cannot scale past one process.** Room codes are unique only within a process; a second instance would hand out colliding codes.
- **Known protocol defects.** `pending.clear()` on any inbound snapshot discards a client's unacknowledged commands; `ackSeq < state.seq` produces spurious `STALE_SEQUENCE` rejections whenever another player moves first; checkpoint restore does not advance the RNG cursor, so post-checkpoint reshuffles diverge.

v2.0 therefore is not a rewrite of the game. It is **the same game with a real product around it**.

1.3 Assumptions recorded at the time of writing

These were decided at kickoff and are recorded so they can be challenged:

1. The client target is **React, web-first, installable as a PWA**, with the Android APK/AAB produced by wrapping the same build (Capacitor). No native-only client.
2. The backend scope is **realtime multiplayer with real accounts**: authenticated identity, persistent progression, server-authoritative matches, and matchmaking.
3. iOS is out of scope for v2.0 but no decision in this document may make iOS impossible later.
4. The five rulesets and their rules are **frozen**. A shedding-type card game's rules are not the subject of this rebuild.

2. Product vision

Color Clash is the card game you can start in four seconds, finish in four minutes, and pick up again on any device you own.

Three sentences that decide arguments:

- **Readability beats spectacle.** At any moment a player must be able to tell whose turn it is, what the active colour is, which of their cards are legal, and what just happened — without colour being the only signal for any of them.
- **The server is the referee.** A client never decides legality, never owns a score, and never authors a balance. This is not an anti-cheat afterthought; it is the reason the architecture exists.
- **Buy a look, never an edge.** Everything purchasable is cosmetic. There is no purchasable draw, wild, extra turn, timer extension or information advantage in any mode.

2.1 Design pillars (carried forward from v1.0, unchanged)

Pillar	Meaning
Instant readability	Turn, active colour, legal moves and consequences are always visible and always have a non-colour cue.
Deterministic rules	Authoritative state is reproducible from seed + command log. The UI never invents game logic.
Modular versions	Five rulesets are separate modules sharing one engine contract.
Fast rounds	Input, animation and resolution feel immediate; latency never creates duplicate actions or divergent hands.
Configuration over hardcoding	Card counts, effects, stacking, player limits and themes live in version manifests.
No hidden rules	Any ambiguity becomes a named flag with a test, never an invented behaviour buried in code.

2.2 What changes in v2.0

Pillar added in v2.0	Meaning
Durable identity	A player is an account, not a string in <code>localStorage</code> . Progress follows the account across devices.
Server-owned truth	Currency, cosmetics, stats and ratings are computed and stored server-side. The client renders them.

Pillar added in v2.0	Meaning
Always someone to play	Matchmaking fills a table in under 15 seconds at target concurrency, and backfills bots rather than making a player wait.
Operable	The service has health checks, metrics, structured logs and a deploy that does not kill live matches.

3. Target audience and personas

3.1 Market position

Color Clash is a **casual competitive shedding card game** on mobile web and Android. Its differentiator is not a new mechanic; it is *five* rulesets that escalate in chaos, sharing one table, one account and one social graph — so a player who has exhausted Classic has somewhere to go without learning a new app.

3.2 Personas

P1 — “Riya, the commuter” (primary, ~60% of sessions) Plays 3–8 minutes at a time, one-handed, on a mid-range Android phone over a patchy 4G connection. Wants a match to start immediately and to survive a tunnel without losing it. Will not read a tutorial. Cares that the app is small and does not drain her battery. *Implications:* cold start budget, reconnect that actually works, offline vs bots, portrait one-handed layout, install size.

P2 — “Arjun, the group-chat host” (primary, drives acquisition) Plays with 3–5 friends in a private room, most often on a call at the same time. He is the one who sends the link. Wants to configure the ruleset, kick the person who went afk, and rematch instantly. *Implications:* shareable room links, robust host controls, rematch that keeps the table, emotes, spectator-free but disconnect-tolerant rooms.

P3 — “Meera, the completionist” (retention) Plays daily, cares about streaks, missions, cosmetics and her rank. Will notice if a stat is wrong. Will spend on a card back she likes. *Implications:* durable stats, season track, achievements, a store, and numbers that are correct because the server computed them.

P4 — “Sam, the sceptic” (edge, but disproportionately loud) Believes the bots cheat and that the shuffle is rigged. Will inspect network traffic. *Implications:* bots receive the identical `PlayerView` a human receives, and this is documented and provable; match seeds are recorded; a match can be replayed from its command log.

3.3 Non-audiences

Tournament organisers, esports, real-money gaming, and under-13 users (the product is rated for 13+ and collects no data that would require a children’s privacy regime).

4. Game content scope

The rules are **frozen from v1.0**. This section exists so the PRD states the product surface, not to re-specify the rules; the authoritative rule text is `RULE_FLAGS.md` plus the version manifests, and the TRD restates the engine contract.

4.1 The five rulesets

Mode	Engine id	Deck	Distinctive mechanics	Complexity
Classic Clash	CLASSIC	108	Skip, Reverse (2-player = Skip), Draw Two, Wild, Wild Draw Four	● ○ ○
Flipstorm	FLIP	112	Dual-sided deck, Flip, Draw One / Draw Five, Skip Everyone, Wild Draw Color	● ● ○
Mayhem	MAYHEM	168	+2/+4/+6/+10 stacking, 25-card elimination, 7 swap, 0 rotate, Discard All	● ● ●
Wild Rush	ALL_WILD	112	Every card structurally wild, global target Draw Two, Double Skip (+3)	● ● ○
Freestyle	FLEX	108	Primary/secondary matrix, public Flex Power token, alternate effects	● ● ●

Requirement PRD-4.1.1 — All five modes ship in v2.0. No mode is gated behind progression, purchase, or account level.

Requirement PRD-4.1.2 — Deck composition is asserted at generation time; a composition whose generated size disagrees with its manifest is a build failure.

Requirement PRD-4.1.3 — Every rule flag (RF-001 ... RF-014) is carried forward with its v1.0 default, and every player-facing rules sheet that presents a flagged default continues to name the flag, so a chosen default is never presented to a player as a law of the game.

4.2 Match configuration

Exposed in the lobby, stored per-room, and validated server-side:

Setting	Values	Default
Win condition	One round · 500 points · Best of three	One round
Draw rule	Draw one · Draw until playable	Draw one
Stacking	On / Off	On for Mayhem, off elsewhere
Wild Draw Four challenge	On / Off	Off (RF-009 — outcome unspecified by source; enabling it raises CHALLENGE_NOT_SPECIFIED)
Clash penalty cards	Integer	2
Elimination threshold	Integer	25 (Mayhem)
Clash grace window	Milliseconds	4000
Scoring	Standard / None	Standard
Table theme	ocean · midnight · ember (+ unlocked)	ocean

Requirement PRD-4.2.1 — Every configuration value is validated server-side against a declared range. A client-supplied value outside its range is rejected, not clamped silently.

4.3 Player counts

- **Rule limit:** 2–10 players. The engine is tested across the full range.
- **Seat limit:** 2–6 players. This is the product limit — what one table can legibly render.

Requirement PRD-4.3.1 — v2.0 keeps the 2–6 seat limit. Raising it is a design change requiring a new table layout, not a configuration change.

5. Feature requirements

Priorities: **P0** = launch blocker · **P1** = required within 60 days of launch · **P2** = planned, may slip.

5.1 Accounts and identity

ID	Priority	Requirement
ACC-01	P0	A player can play a full local match against bots with no account at all . Nothing about account creation blocks first play.
ACC-02	P0	A guest identity is created on first launch and is a real server-side account row, so progress made before signing in is not lost.
ACC-03	P0	A guest can upgrade to a full account (email + password, Google Sign-In) without losing coins, cosmetics, stats or rating.
ACC-04	P0	Sign-in is required for online play. Guest accounts may play online; anonymous unauthenticated sockets may not.
ACC-05	P0	An account's display name is 1–16 characters, unique-per-account but not globally unique ; players are disambiguated by a short discriminator.
ACC-06	P0	Display names are filtered against a profanity list at set time, and a name can be reset by moderation.
ACC-07	P0	Sessions are refreshable; a returning player is not asked to sign in again for at least 30 days of activity.
ACC-08	P1	Account deletion is self-service and removes or irreversibly anonymises all personal data within 30 days.
ACC-09	P1	A player may be signed in on multiple devices; a second device joining an active match takes over the seat and the first is told why.
ACC-10	P2	Apple Sign-In (required before any iOS release).

5.2 Play modes

ID	Priority	Requirement
PLAY-01	P0	Quick Play (offline) — instant match vs bots, no network, any ruleset, 2–6 seats, configurable bot tiers.
PLAY-02	P0	Pass-and-play — 2–6 human seats on one device, hands hidden between turns behind a confirm step.
PLAY-03	P0	Private room — host creates a room, shares a 5-character code <i>and</i> a deep link; up to 6 seats; empty seats fillable with bots at any tier.
PLAY-04	P0	Quick Match (online) — a matchmaking queue per ruleset that produces a table of real players, backfilled with bots if the queue does not fill within the SLA.
PLAY-05	P1	Ranked — a rated queue for Classic Clash only at launch, with a visible rank tier and a season reset.
PLAY-06	P1	Resume — an interrupted local match is resumable from the menu; an interrupted online match is rejoinable while the room lives.
PLAY-07	P2	Friends — add by code, see who is online, invite directly into a room.

ID	Priority	Requirement
PLAY-08	P2	Tournaments / events — time-limited bracket or challenge events.

Requirement PLAY-04.1 — Matchmaking SLA: a queued player is seated within **15 seconds** at target concurrency. If a real table cannot be formed within **30 seconds**, the queue seats the player against bots rather than continuing to wait, and tells them that is what happened.

5.3 In-match experience

ID	Priority	Requirement
MATCH-01	P0	The client never computes legality. Legal card ids arrive from the authoritative engine.
MATCH-02	P0	A turn clock, owned and broadcast by the server as an absolute deadline, with a default allowance of 30 seconds per decision.
MATCH-03	P0	Disabled in local play. On timeout the server plays the least presumptuous legal action on the player's behalf: answer a pending choice with the first eligible option; else pass if passing is legal; else draw and pass; else play the first legal card.
MATCH-04	P0	Three consecutive timeouts convert the seat to a bot of the same difficulty for the rest of the match, and the player is told.
MATCH-05	P0	Disconnection holds the seat. The room survives while at least one human is connected, and is held for 90 seconds after the last human drops.
MATCH-06	P0	On reconnect a client receives a full snapshot plus every missed event, and any in-flight command is replayed under its original id and applied exactly once.
MATCH-07	P0	The Clash call: a hand going from more than one card to exactly one enters a pending state with a 4-second grace window; missing it before the next game action costs 2 cards. Repeated calls are idempotent.
MATCH-08	P0	Emotes — a closed, server-validated vocabulary of 8, rate-limited per player.
MATCH-09	P1	Free-text chat in private rooms only, filtered, rate-limited, reportable, and off by default for accounts under 18.
MATCH-10	P1	Rematch keeps the table: same seats, same ruleset, new seed, one tap.
MATCH-11	P2	Match replay from the stored command log, scrubable, shareable by link.

5.4 Bots

ID	Priority	Requirement
BOT-01	P0	Four tiers: Easy, Normal, Hard, Chaos.
BOT-02	P0	A bot receives exactly the <code>PlayerView</code> a human client receives. There is no code path from a bot to authoritative state. Difficulty may only improve decision quality, never create information.
BOT-03	P0	Bots submit the same commands, through the same validation pipeline, as humans — including rate limiting and duplicate detection. (v1.0 bypassed all three; this is a defect being fixed.)

ID	Priority	Requirement
BOT-04	P0	Easy and Normal bots <i>miss</i> Clash calls at a tuned rate (45% / 15%), because the cost is the ordinary 2-card penalty. This is modelled as decision quality, not as cheating.
BOT-05	P1	Bots that backfill an online table are visually indistinguishable in skill from a weak human but are always labelled as bots in the seat UI.

5.5 Progression, economy and store

ID	Priority	Requirement
ECON-01	P0	All balances are server-authoritative. The client displays; it never computes or persists a balance.
ECON-02	P0	Two currencies: Coins (soft, earned from play) and Gems (hard, purchased or granted). Neither buys a gameplay advantage in any mode.
ECON-03	P0	New accounts start at 0 coins / 0 gems and level 1. (v1.0 seeded 12,450 / 860 client-side; those balances are not migrated.)
ECON-04	P0	Coin grant per completed match: a base award plus a win bonus, both server-computed and both capped per rolling 24 hours to make farming pointless.
ECON-05	P0	Cosmetic inventory: card backs, table themes, avatars, profile frames, emote packs, sound packs. All are cosmetic and none alter engine input or output.
ECON-06	P1	A store with coin-priced and gem-priced cosmetics, and gem bundles as real-money IAP.
ECON-07	P1	Daily missions and a 7-day login streak, both server-evaluated.
ECON-08	P1	A cosmetic-only season track with a free and a paid lane.
ECON-09	P0	Explicitly prohibited: purchasable draws, wilds, extra turns, timer extensions, rerolls of a hand, previews of an opponent's hand, or any purchase that changes a Command, a Resolution or an RNG draw.
ECON-10	P1	Optional rewarded video ads in solo/offline contexts only. Never during an online match, never mid-turn, never as a condition of continuing.
ECON-11	P0	Every balance change is written to an append-only ledger with a reason code. A balance that cannot be reconstructed from its ledger is a Sev-2 incident.

5.6 Stats, leaderboards and rating

ID	Priority	Requirement
STAT-01	P0	Per-account, per-ruleset lifetime stats: matches played, won, win rate, longest win streak, cards played, Clash calls made and missed.
STAT-02	P0	Match history — last 50 matches, each with ruleset, final standings, duration and date.
STAT-03	P1	A rating for ranked Classic Clash (Glicko-2 style), hidden as a number, surfaced as a tier.

ID	Priority	Requirement
STAT-04	P1	Leaderboards: global and friends, weekly and all-time, per ruleset, recomputed on a schedule rather than on read.
STAT-05	P0	Stats are derived only from server-recorded match results. A client-reported result is never trusted.

5.7 Onboarding and learning

ID	Priority	Requirement
LEARN-01	P0	First launch reaches a playable table in under 20 seconds including naming the player.
LEARN-02	P0	A per-ruleset rules sheet, reachable from the mode tile, the lobby and the in-match pause sheet.
LEARN-03	P0	Contextual first-time hints for each ruleset's distinctive mechanic (Flip, stacking, the Flex token, target selection), shown once and dismissible forever.
LEARN-04	P1	A scripted 90-second interactive tutorial for Classic Clash, replayable.
LEARN-05	P0	An illegal tap is explained in words, not just refused — “Match colour, number, or play a wild”.

5.8 Settings and accessibility

ID	Priority	Requirement
A11Y-01	P0	Colour is never the only signal. Every card carries a text glyph and an <code>aria-label</code> ; the active colour is named in text beneath the pile.
A11Y-02	P0	Reduced motion honours the OS setting and an in-app override, and the override wins.
A11Y-03	P0	Every colour/ink pairing in the palette clears WCAG AA contrast, asserted by a test that fails the build.
A11Y-04	P0	Touch targets on the table are at least 44x44 CSS pixels.
A11Y-05	P0	Sound and haptics toggles; “off” means no audio node is created.
A11Y-06	P1	Left-handed layout, larger text, and a high-contrast table theme.
A11Y-07	P1	Screen reader support for the table: live region for the event log, and announced turn changes.
A11Y-08	P0	The game is fully playable in portrait and in landscape on a phone, and in a desktop browser window.

5.9 Platform and distribution

ID	Priority	Requirement
DIST-01	P0	Installable PWA: offline-capable for local play, precaching every emitted asset, never intercepting cross-origin requests or WebSockets.

ID	Priority	Requirement
DIST-02	P0	Signed Android AAB + APK built in CI from encrypted keystore secrets, with <code>compileSdk/targetSdk</code> pinned explicitly rather than inherited.
DIST-03	P0	Play Store listing complete: privacy policy, Data Safety declaration, content rating, and an accurate description of what the online mode transmits.
DIST-04	P1	Deep links — <code>colorclash.app/r/<CODE></code> opens the installed app directly into the room, or the web app if not installed.
DIST-05	P2	iOS via the same Capacitor shell.

6. Success metrics

6.1 Launch gates (must hold before general availability)

Metric	Target
Crash-free sessions	≥ 99.5%
Cold start to interactive menu (mid-range Android, 4G)	≤ 2.5 s p75
Time to first playable table, new install	≤ 20 s p75
Matchmaking time to seated	≤ 15 s p75, ≤ 30 s p99
Command → authoritative acknowledgement	≤ 150 ms p50, ≤ 400 ms p95 (same region)
Desync incidents per 1,000 matches	0
Match completion rate (started → resolved, online)	≥ 92%

6.2 Product health (reviewed weekly)

Metric	Target at 90 days
D1 / D7 / D30 retention	40% / 18% / 8%
Median session length	9 minutes
Matches per daily active user	≥ 3.5
Share of sessions that include an online match	≥ 35%
Guest → registered conversion	≥ 25% within 7 days
Paying conversion	≥ 1.5%
ARPPDAU	Tracked, no target at launch
Reconnect success rate (dropped socket → resumed same match)	≥ 95%

6.3 Counter-metrics

Metrics that must **not** move in the wrong direction while the above improve:

- Median turn duration (a rise means the UI got slower or more confusing).
- Rate of rejected commands per match (a rise means a protocol regression).
- Share of matches ending in a timeout-converted bot seat.
- Support contacts about missing currency or missing progress.

7. Analytics

Requirement ANA-01 — Analytics run behind a provider abstraction, so the vendor can be replaced without touching call sites.

Requirement ANA-02 — No analytics event fires before a privacy consent decision is recorded. Declining consent disables collection but never disables gameplay.

Requirement ANA-03 — No event carries a hand, a card id in a private hand, an email address, or an IP address.

Minimum event set, carried forward from v1.0 and extended:

Event	Purpose
app_opened, first_open	Install funnel
account_created, guest_upgraded, sign_in	Identity funnel
match_started	Mode / ruleset funnel; carries ruleset, seat count, bot count, source (local/room/queue)
turn_completed	Pacing and drop-off; carries duration
card_played	Rule engagement and balance
clash_called, clash_missed	UX clarity of the Clash window
match_completed, match_abandoned	Retention signal and completion rate
matchmaking_queued, matchmaking_seated,	Queue health
matchmaking_bot_backfill	
reconnect_attempted, reconnect_succeeded	Network reliability
command_rejected	Protocol regression detector; carries reason code
desync_detected	Sev-1 alarm
tutorial_completed, hint_dismissed	Learning effectiveness
store_viewed, purchase_started, purchase_completed,	Revenue funnel
purchase_failed	
ad_offered, ad_completed	Ad engagement

8. Legal, privacy and content policy

ID	Requirement
LEG-01	RF-014 remains closed. No third-party artwork, audio, typography, layout or wordmark ships in any build. The CI branding audit runs on every push and fails the build if the prior mark reappears.
LEG-02	The only third-party assets in the product are the Barlow and Nunito typefaces under SIL OFL 1.1, recorded in docs/BRANDING.md with source, licence and attribution requirements.
LEG-03	Before any external asset is added, its source, licence, commercial-use permission and attribution requirement is recorded in that table. An asset that cannot be described that way does not ship.
LEG-04	The branding audit is a regex and cannot look at an image or judge similarity. Any new binary asset requires human review; this is stated so the audit is not mistaken for a guarantee.
LEG-05	A published privacy policy covers: what is collected (account identifier, display name, gameplay telemetry), what is not (contacts, location, advertising identifiers unless ads are enabled and consented), retention, and deletion.
LEG-06	Age gate at account creation. Accounts declaring under 13 are refused. Accounts under 18 have chat off by default and are excluded from ad personalisation.
LEG-07	Player-generated content is limited to display names and free-text chat in private rooms. Both are filtered and reportable.
LEG-08	No loot boxes, no randomised paid rewards, no real-money wagering, no currency cash-out.
LEG-09	Play Store Data Safety declarations must match the implementation and are re-verified at every release that changes data handling.

9. Release plan

9.1 Phasing

Phase	Scope	Exit criteria
0 — Foundation	New monorepo scaffold, CI green on every gate (lint, format, both typecheck modes, unit, e2e, branding audit, all three build targets). Engine and content packages ported unchanged.	Every CI gate executes and passes. This closes the v1.0 gap where four gates had never run anywhere.
1 — Identity & persistence	Auth service, account model, database, guest accounts, guest upgrade, server-owned profile and inventory.	A player can sign in on two devices and see the same profile.
2 — Authoritative match service	Rebuilt match server: real WebSocket library, per-connection auth, shared room registry, fixed ack/resync semantics, RNG cursor in the persisted room.	Two clients survive a server restart mid-match without divergence.
3 — Rooms & social	Private rooms with codes and deep links, host controls with real authorisation, emotes, rematch.	Arjun's flow end to end on four devices.
4 — Matchmaking	Queue service, bot backfill, seating SLA.	SLA met under synthetic load at target concurrency.
5 — Progression & economy	Ledger, currencies, inventory, store, missions, streaks.	Every balance reconstructible from its ledger.
6 — Stats, history & leaderboards	Match result recording, stats rollups, leaderboards, ranked rating.	Stats agree with recomputation from raw match rows.
7 — Polish & accessibility	Animation pass, motion budgets, screen reader, high contrast, left-handed layout, onboarding hints, tutorial.	Accessibility audit passes; three-viewport e2e green.
8 — Hardening & launch	Load test, chaos test (kill a node mid-match), security review, store submission, runbooks.	Launch gate metrics in §6.1 met on a staging environment.

9.2 Out of scope for v2.0

Spectator mode · voice chat · cross-play with any other product · real-money wagering · user-created rulesets · clans/guilds · iOS release · localisation beyond English (though all player-facing strings must be externalised so it is a content task, not an engineering one).

10. Risks

#	Risk	Impact	Mitigation
R1	Porting the engine introduces a subtle rule regression.	High	The engine and content packages move unmodified , with their 212 tests as the acceptance gate. No rule change is in scope.
R2	The four CI gates that have never run (lint, format:check, typecheck:installed, the Vite builds) fail extensively on first execution.	Medium	Phase 0 exists solely to run them. The honest expectation is that typecheck:installed <i>will</i> find real problems, because the offline types/ shims are narrower than the real definitions and have accepted code they will reject.
R3	Matchmaking has nobody to match at low concurrency.	High	Bot backfill at 30 s is a launch requirement, not a fallback. Regional queues are merged below a concurrency floor.
R4	Migrating v1.0 localStorage balances creates an unearned economy.	Medium	Balances are not migrated. v1.0's 12,450 coins were a placeholder, never earned. A one-time cosmetic grant acknowledges early players instead.

#	Risk	Impact	Mitigation
R5	The hand-rolled WebSocket implementation's known limits (no frame size cap, no backpressure, no heartbeat) become a live DoS.	High	Replace it in Phase 2 with a maintained library. The <code>WsConnection</code> interface already isolates it, which is why this is a swap and not a rewrite.
R6	Server-authoritative economy increases backend cost per match beyond what a casual title supports.	Medium	Ledger writes are batched at match end, not per turn; stats are rolled up on a schedule, not on read.
R7	Trademark or similarity challenge despite the rebrand.	High	RF-014 closed with original generated assets and a CI audit; legal review of the store listing before submission; documented asset provenance.
R8	Android review rejects on Data Safety or content rating mismatch.	Medium	Data Safety form completed from the actual implementation as a release checklist item, re-verified whenever data handling changes.

11. Open questions

#	Question	Needed by
Q1	Do v1.0 players get any recognition grant, and in what form?	Phase 5
Q2	Is ranked launched with Classic Clash only, or all five rulesets?	Phase 6
Q3	Gem bundle price points and regional pricing.	Phase 5
Q4	Is free-text chat worth its moderation cost at this scale, or do emotes suffice?	Phase 3
Q5	Which region does the first match server cluster live in, and what is the second?	Phase 4
Q6	Ads: launch with them, or hold until retention is proven?	Phase 5

12. Acceptance criteria for v2.0

The rebuild is done when all of the following are true.

1. A brand-new device reaches a playable table in under 20 seconds with no account.
2. That player can sign in later and keep everything they earned.
3. Four players on four devices, on different networks, play a full Mayhem match; one of them force-quits mid-stack and rejoins to the correct state.
4. A server node is killed mid-match; the match resumes on another node with no divergence in hands, piles or turn order.
5. Every currency balance in the database can be reconstructed by replaying its ledger.
6. A bot cannot be shown to have used information a human in the same seat did not have, and the code path that would allow it does not exist.
7. Every CI gate — lint, format, both typecheck modes, unit, e2e at three viewports, branding audit, and all three build targets — runs and passes on every push.
8. All five rulesets play a complete match, and the full v1.0 edge-case suite passes unmodified.
9. The signed AAB installs from the Play internal testing track and plays online against the web client.
10. No build contains any third-party asset outside the two OFL typefaces.